AI 101 for Mechanical Engineers · Module 1

Automation, robots and general intelligence

Lesson 1.2

What you will learn

> Distinguish automation, robotics and learning, and explain narrow AI and the proposed idea of artificial general intelligence (AGI).

Lesson overview

Module 1 · Lesson 2

Automation, robots and general intelligence

AI 101 for Mechanical Engineers

1 Automation and robotics

Automation carries out a process with less manual intervention.

Measured temperature → Fixed rule → Switch off above 30 °C

Teaching example only — not an equipment setting.

A controller can follow fixed rules, such as PID feedback, without learning from data.

Robotics

A robot senses, decides and acts. It can use conventional control, AI or both.

The word robot does not tell you whether ML is involved.

2 Narrow AI and AGI

Narrow AI

Designed for particular tasks.

Example | A classifier identifies a defined set of defects.

Artificial general intelligence (AGI)

Proposed capabilities across many tasks.

People still disagree about how to define and test AGI.

Doing one task well does not prove broad capability.

3 Ask five questions

1. **Inputs:** What information does it receive?
2. **Learned components:** Which parts learn from data?
3. **Outputs:** What does each part produce?
4. **Permitted actions:** What is it allowed to do?
5. **Failure consequences:** What happens if it is wrong?

Bearing alert → Qualified review

An alert is a reason to investigate, not a confirmed diagnosis.

A well-written explanation is not enough to approve a pressure-vessel design.

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Automation means less manual intervention

- A controller can follow fixed rules [6]
- Example: a thermostat switches off above 30 °C
- This is automation, not learning from examples
- 30 °C is a teaching value, not a setting

A robot senses, decides and acts

- An introductory simplification; see ISO 8373 [89]
- Decisions may come from control rules, AI or both
- A PID controller follows feedback rules [7]
- “Robot” does not tell you whether it learns

Narrow AI is judged task by task

Narrow AI

- Designed for particular tasks
- Evidence covers the tasks tested

AGI

- Proposed broad capability
- Definitions and tests disputed [8]

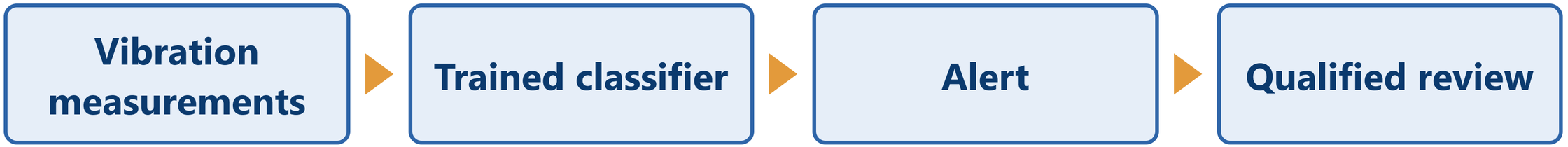
> Fluent text does not give authority to approve a pressure-vessel design.

Five questions for any system

1. Inputs: what information does it receive?
2. Learned components: which parts learn from data?
3. Outputs: what does each part produce?
4. Permitted actions: what may it do?
5. Failure consequences: what if it is wrong?

> These questions reveal details a product label may leave out [9].

A bearing alert is a reason to investigate



- Not a confirmed diagnosis or automatic shutdown
- Check missed faults and false alarms first (Module 8)
- Check performance on representative examples [10]

Match each claim with its evidence

- Fixed controller: follows a specified rule
- ML component: show that it actually learns
- Classifier: performance evidence for its task
- AGI claim: clear definition and suitable tests

CHECK YOUR UNDERSTANDING

Is a conventional PID-controlled robot necessarily using ML?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Is a conventional PID-controlled robot necessarily using ML?

> No. It can follow specified feedback rules without learning. If another part learns, identify it separately.

CHECK YOUR UNDERSTANDING

Does a thermostat switching off above 30 °C demonstrate AGI?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does a thermostat switching off above 30 °C demonstrate AGI?

> No. It follows a fixed automation rule and shows no broad general capability.

Key takeaways

- Automation reduces manual intervention
- Robots sense, decide and act, with or without ML
- Narrow AI handles defined tasks; AGI is disputed
- Check inputs, learning, outputs, actions and failures

References

[6] R. Parasuraman, T. B. Sheridan, and C. D. Wickens, “A model for types and levels of human interaction with automation,” IEEE Trans. Syst., Man, Cybern. A, Syst. Humans, vol. 30, no. 3, pp. 286–297, 2000. doi: 10.1109/3468.844354

[7] K. J. Åström and R. M. Murray, Feedback Systems: An Introduction for Scientists and Engineers, 2nd ed. Princeton, NJ, USA: Princeton Univ. Press, 2021.

[8] M. R. Morris et al., “Position: Levels of AGI for operationalizing progress on the path to AGI,” in Proc. 41st Int. Conf. Mach. Learn. (ICML), PMLR, vol. 235, 2024, pp. 36308–36321.

[9] National Institute of Standards and Technology, “Artificial Intelligence Risk Management Framework (AI RMF 1.0),” NIST AI 100-1, Gaithersburg, MD, USA, 2023. doi: 10.6028/NIST.AI.100-1

[10] Y. Lei, B. Yang, X. Jiang, F. Jia, N. Li, and A. K. Nandi, “Applications of machine learning to machine fault diagnosis: A review and roadmap,” Mech. Syst. Signal Process., vol. 138, Art. no. 106587, 2020. doi: 10.1016/j.ymssp.2019.106587

[89] Robotics — Vocabulary, ISO 8373:2021, 3rd ed., International Organization for Standardization, Geneva, Switzerland, Nov. 2021.

Numbers match the course reference list. The written lesson is the authoritative text.